

Types include DRAM (Dynamic RAM, needs refreshing) and SRAM (static RAM, faster but more expensive).

Example: Running apps, open files, or browser tabs are stored in RAM.

⑥ ROM (Read-Only Memory)

> Purpose: It stores firmware or permanent instructions, like the BIOS, needed to boot the system.

> Characteristics:

- Non volatile: Retains data without power.
- Read-only in most cases, though some types (e.g., EEPROM) allow limited writing.

Example: Firmware for starting the operating system.

• Secondary Memory

↳ Secondary memory is non-volatile storage used for long-term data retention. It is slower than primary memory but offers much larger storage capacity and is used to store data and programs not currently in use.

↳ Types of Secondary Memory

① Hard Disk Drives (HDD):

> Purpose: It stores large amount of data, including the operating system, applications, and user files.

> characteristics:

- Non-Volatile, slower than SSDs, mechanical parts (spinning disks).

- High capacity (e.g. 1TB, 4TB).

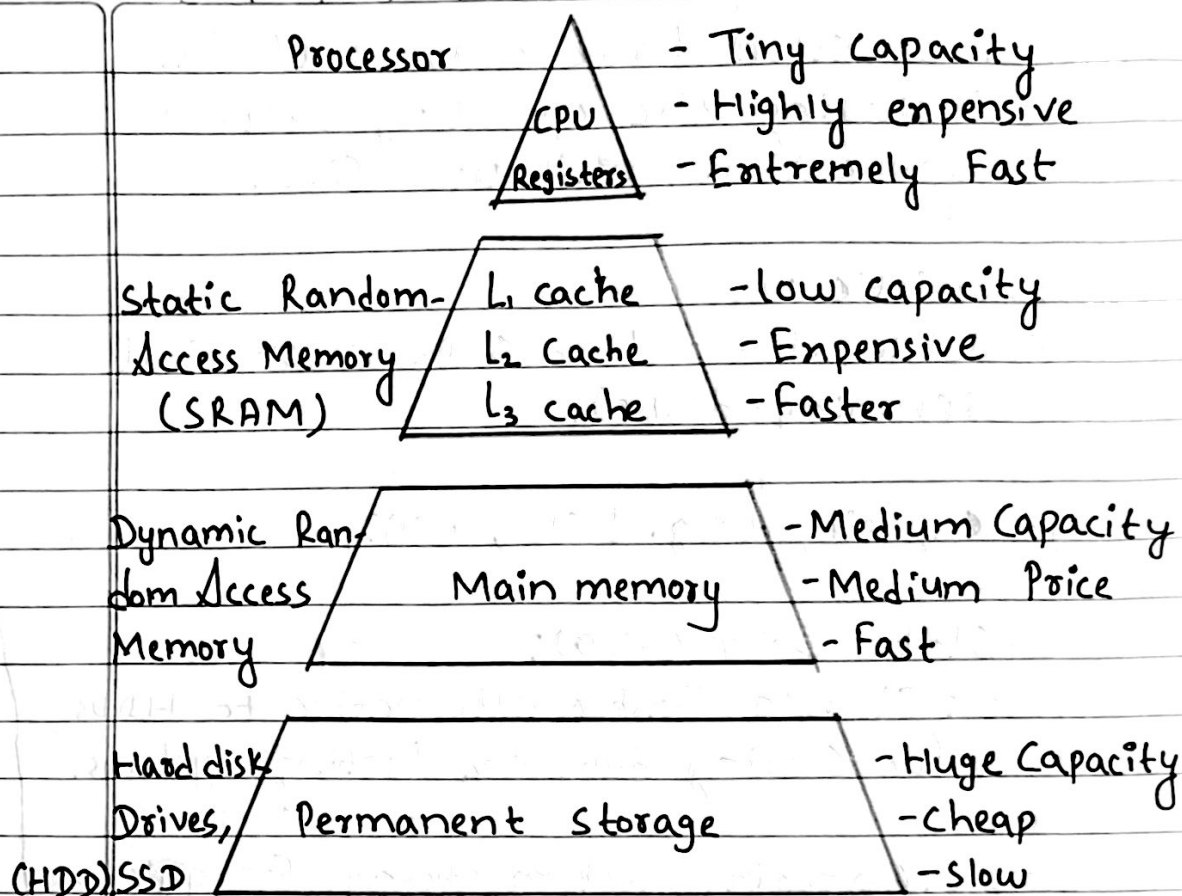
⑥ Solid-State Drive (SSD):

> Purpose: It is a Faster alternative to HDDs for storing data and booting systems.

Example: Operating system storage for quicker boot times.

Memory Hierarchy

↳ Memory hierarchy in a computer system refers to the arrangement of storage devices in a hierarchy based on speed, cost and capacity. The faster and more expensive the memory the closer it is to the CPU. Here's the basic structure of the memory hierarchy:



1. Registers

- **Location:** Inside the CPU (Central Processing Unit)
- **Speed:** The fastest Memory in a computer system. It operates at the same speed as the CPU's clock.
- **Cost:** Extremely expensive on a per-bit basis due to their high speed and small size.

- **Capacity:** Smallest in capacity, usually only a few dozen to a few hundred bytes.
- **Function:** Registers hold data that is immediately needed by the CPU during execution. Examples include the instruction registers, ~~which are~~ accumulator, and data registers. Operations are performed directly on data stored in the registers, which avoids delays associated with fetching data from slower memories.
- **Uses:** Used for temporary storage of operands, intermediate results, and addresses that frequently accessed or being processed by the CPU.
- **Types of registers**
 - ① **Accumulate (ACC):** It stores results of arithmetic / Logical operations.
 - ② **Program Counter (PC):** It Holds the address of the next instruction to execute.
 - ③ **Memory Address Register (MAR):** It stores address of Memory to read/write.

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④ Instruction Register (IR): It Holds the current instruction being decoded / executed.

⑤ Memory Buffer Register (MBR) / Memory Data Register (MDR)

: Temporarily stores data read from or written to memory.

2. Cache Memory

- Location: Positioned between the CPU and main memory, either on the CPU chip (L1 and Sometimes L2 cache) or on a separate chip (L2, L3 cache).
- Speed: Faster than RAM but slower than registers.
- Cost: More expensive than RAM ~~and~~ but cheaper than register.
- Capacity: Larger than registers but Smaller than RAM. It typically range from a few kilobytes (KB) for L1 cache to several megabytes (MB) for L3 cache.
- Whenever the data requested by the processor is found in cache memory, it is known as cache hit. Otherwise it is known as cache miss.

- **Function:** Cache memory stores copies of data that is frequently used by the CPU, reducing the time it takes to access data compared to fetching it from RAM.
 - > **L1 cache** - The closest to the CPU and the smallest, but the fastest. Usually located on the CPU chip itself.
 - > **L2 cache** - Larger and slower than L1 but still fast. It may also be located on the CPU or on a separate chip.
 - > **L3 cache** - The largest and slowest cache level, shared between multiple CPU cores in many modern processors.
- **Usage** - Helps bridge the speed gap between the fast CPU and slower main memory (RAM).

3. Main memory (RAM).

- **Location** - Connected to the motherboard and directly accessible by the CPU.
- **Speed** - Slower than cache memory but significantly faster than secondary storage.
- **Cost** - less expensive than cache and registers,

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but more expensive than secondary storage.

- **Capacity** - Typically ranges from a few gigabytes (GB) to several Terabytes (TB) in modern computers.
- **Function** - RAM (Random Access Memory) is volatile, meaning ~~that~~ that it loses its data when the power is turned off. It stores the operating system, application software, and currently active data so that the CPU can access it quickly.
 - **Dynamic RAM (DRAM)** - The most common types of RAM used in modern systems.
 - **Static RAM (SRAM)** - Faster but more expensive than DRAM, used in ~~the~~ caches.
- **Usage** - Acts as the working memory of the computer, providing a place for the CPU to read and write data that is actively being processed.

4. Secondary Storage (Hard Drives, SSDs)

- Location - Outside the CPU, connected via interfaces like SATA, NV Me, etc.
- Speed - Much slower than RAM, HDDs (Hard Disk Drives) have mechanical components that cause delays, while SSDs (Solid State Drives) are faster but still slower than RAM.
- Cost - Cheaper than RAM on a per-bit basis. SSDs are more expensive than HDDs but provide better performance.
- Capacity - Typically much larger than RAM, with capacities ranging from hundreds of gigabytes (GB) to several Terabytes (TB).
- Function - Secondary storage is non-volatile, meaning that it retains data even when the computer is powered off. It is used for long term storage of files, applications, and the operating system.

➤ Hard Disk Drives (HDDs) - Use magnetic storage to read and write data. They are slower due to mechanical movement but offer large storage at a lower cost.

> Solid State Drives (SSDs) - Use flash memory, which is faster and more reliable than HDDs because there are no moving parts. SSDs are increasingly common for performance gains.

- Usage - stores files, software, and backups. The operating system and applications are loaded from secondary storage into RAM when the computer is turned on or when a program is run.

5. Tertiary Storage and off-Line Storage (Optical Discs, Magnetic Tapes)

- Locations - External to the system; usually connected through specialized drives.
- Speed - Much slower than both main memory and secondary storage.
- Cost - Cheapest in terms of cost per bit of data storage.
- Storage - Tertiary storage devices can ~~storage~~ store large amounts of data, typically up to several terabytes or more.
- Function - Tertiary storage is used primarily for backups, archives, and data

retrieval when needed. It is often external and accessed infrequently. Examples include:

- > Optical Discs (CDs, DVDs, Blu-rays)
 - Used for media storage, backups, and software distribution. They have limited rewritability and relatively slow access times.
- > Magnetic Tapes
 - Used in enterprise environments for backups and archiving large data-sets. They have very high capacity ~~and~~ but slow sequential access times.
- Usage - Ideal for long-term data storage and backup that doesn't need to be accessed frequently. Often used by enterprises for disaster recovery, archival purposes, and in some cases, large-scale data migration.

* Types of RAM (Random Access Memory)

- RAM is the main memory of a computer and can be categorized into two main types

• ~~DRAM~~ DRAM (Dynamic RAM)

- > Speed - Slower than SRAM
- > Cost - Cheaper than SRAM
- > Capacity - Larger capacity
- > Technology - Requires periodic refreshing to maintain data, as each bit is stored in a capacitor that leaks charge.
- > Usage - Used as main memory in computers. Modern systems use variations of DRAM like DDR (Double data Rate) RAM (e.g. DDR4, DDR5).

• SRAM (Static RAM)

- > Speed - Faster than DRAM
- > Cost - More expensive
- > Capacity - Smaller capacity compared to DRAM
- > Technology - Does not require refreshing as it uses flip-flops to store each bit, making it faster and more reliable.
- > Usage :- Used for cache memory (L1, L2, and L3) and in small, fast memory banks within the CPU.

* Types of ROM (Read-only Memory)

- ROM is non-volatile memory, meaning it retains data even when power is off. It is used for storing firmware and other permanent instructions.

- PROM (Programmable ROM)
 - > Programmability - Can be programmed once after manufacturing using a special device.
 - > Usage - Used in situations where data must be written after manufacturing but not changed afterward.
- EPROM (Erasable Programmable ROM)
 - > Rewritability - Can be erased by exposing it to ultraviolet light and then programmed.
 - > Usage - Used in situations where firmware might need updates, but not frequently.
- EEPROM (Electrically Erasable Programmable ROM)
 - > Rewritability - Can be erased and reprogrammed electrically, without needing special equipment.
 - > Usage - Used in systems where data updates are required but less frequently than RAM, such as BIOS firmware.
- Flash Memory
 - > Rewritability - A type of EEPROM, but faster and more efficient. It is

often used for storage in USB drives, SSDs, and other devices.

> Usage - Allows multiple rewrites, making it useful for modern systems requiring both storage and quick data access, such as mobile devices.

* Types of Magnetic Storage devices

- Magnetic storage devices are a type of non-volatile storage medium that use magnetic fields to store and retrieve data. Here are some common types:

• HDDs (Hard Disk Drives)

> Description - HDDs are widely used in computers and other electronic devices. They consist of one or more spinning disks (Platters) coated with magnetic material.

> Use - Commonly used in desktops, laptops, and servers for storing large amounts of data.

• Magnetic Tape Drives

> Description - Magnetic Tape ~~Dr~~ is a thin strip of plastic material coated

* Magnetic storage device use Read, Write, Head
electromagnetic induction ← Electromagnet

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With a magnetic layer, typically made of iron oxide or other magnetic materials. The data is recorded by magnetizing tiny portions of the tape in different directions. Data is written and read sequentially.

> Use - Used primarily for archival and backup purposes due to their high capacity and low cost per byte.

• Floppy Disks (Storage - 1.44 MB)

> Description - A now-obsolete magnetic storage device that was widely used from the 1970s to the early 2000s. They come in various sizes, with 3.5-inch and 5.25-inch disks being the most common.

> Use - Historically used for personal computing and transferring small files.

• Zip Drives

> Description - A removable floppy disks storage system introduced by Iomega in the 1990s. It provided higher storage capacity than traditional floppy disks.

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> Use - Used for personal data storage and backups, now largely obsolete.

- Magnetic stripe Cards

> Description - These are used in credit cards, debit cards, and ID cards. The Magnetic Stripe stores data that can be read by swiping the cards through a reader.

> Use - Commonly used for access control, financial transactions, and identification.

- * Types of Optical Storage devices

↳ Data read & write in the form of Pits & Land ↑ burns
↓ flat surface

↳ Optical drives are devices used to read and write data to optical discs using laser technology. These discs include CDs, DVDs, and Blu-rays. Here are the main types of optical drives.

- ① CD-ROM (Compact Disc Read-Only Memory) Drives

> Description - CD-ROM drives are designed to read data from CDs, which can store up to 700 MB of data. These drives cannot write data; they can only read pre-recorded CDs.

> Use - Commonly used for reading software, music, and other digital content distributed on CDs.

② CD-RW (Compact Disc - ~~Reor~~ Rewritable) Drives

> Description - CD-RW drives can write and rewrite data to CD-RW discs multiple times. These drives offer more flexibility than CD-R drives.

> Use - Ideal for tasks that require frequent updating of data, such as backups and temporary storage.

③ CD-R (Compact Disc - Recordable) Drives

> Description - CD-R drives allow users to write data to blank CD-R discs. Once data is written, it cannot be erased or modified.

* CD - 700MB

> Use - Used for creating personal music CDs, data backups, and software distribution.

④ DVD-ROM (Digital Versatile Disc Read-Only Memory) Drives

> Description - DVD-ROM drives are designed to read data from DVDs, whi-

ch typically hold 4.7 GB of data (single-layer) or 8.5 GB (dual-layer).

> Use - Used for reading DVD movies, software, and data distributed on DVDs.

⑤ DVD-R (Digital Versatile Disc-Recordable) Drives

> Description - DVD-R Drives can write data to blank DVD-R discs. These discs are write once, meaning the data cannot be altered after it is recorded.

> Use - Used for creating personal video discs, software backups, and large data storage.

⑥ DVD-RW (Digital Versatile Disc-Rewritable) ~~Device~~ Drives

> Description - DVD-RW drives allow data to be written, erased, and rewritten to DVD-RW discs. This offers flexibility similar to CD-RW but with much higher storage capacity.

> Use - Suitable for tasks that require updating large amounts of data or creating temporary backups.

① Blu-ray Disc (BD) Drives

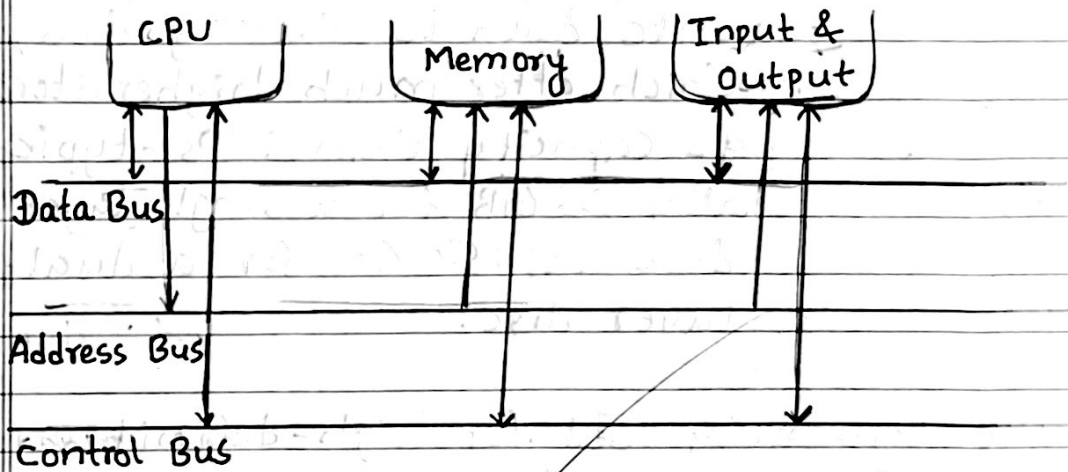
- > Description - Blu-ray Drives can read and write data to Blu-ray discs, which offer much higher storage capacity than DVDs - typically 25 GB for a single-layer disc and 50 GB for a dual-layer disc.
- > Use - Commonly used for high-definition video storage, large data backups, and Gaming.

Bus and its types

- ↳ Buses, in the context of computer ~~arch~~ architecture and digital communication, refer to the communication systems that transfer data between components inside a computer or between computers. A bus is essentially a set of physical connections like cables or printed circuits through which data is transmitted.

It acts as a common pathway that connects multiple hardware components, allowing them to communicate with each other. The bus facilitates the movement of data, address, control signals, and power between the

processor (CPU), memory, input/output (I/O) devices, and other peripheral components.



• Data Bus

↳ A Data Bus is a communication pathway that is ~~transfer~~ used to transfer actual data between the various components of a computer, such as the CPU, memory (RAM), and input/output (I/O) devices. The data bus carries binary data (0s and 1s), which can represent instructions or information to be processed.

> Key features of Data Bus:

① Bi-Directional

- Data can flow in both directions, meaning it can be transferred to and from the CPU, memory, and other peripherals.

② Width (Bus-width)

- The width of ~~bus~~ the data bus (measured in bits) determines how much data can be

transferred at one time. For example, a 32-bit data bus can transfer 32 bits of data simultaneously.

- Common data bus widths include 8-bit, 16 bit, 32 bit, 64 bit. The wider the bus, the more data it can transfer in a single cycle, which improves performance.

③ Speed (Bus Speed)

- The speed at which the data bus operates determines how fast data can be transferred. This is typically measured in megahertz (MHz) or gigahertz (GHz).

• Data Bus in Action:

➤ Reading Data - When the CPU reads data from memory or an I/O device, the data is sent through the data bus to the CPU for processing.

➤ Writing Data - When the CPU writes data to memory or a device, it sends the data through the ~~bus~~ data bus to the target location.

• Address Bus

➤ An address bus is a computer bus that is used to transfer memory addresses from the

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CPU to other components like RAM and Input/Output (I/O) device. The main purpose of the address bus is to specify the location (address) in memory where data is to be read from or written to.

• Key features of the Address Bus:

1. Unidirectional

- Unlike the data bus, the address bus is typically unidirectional, meaning that it only carries information in one direction - from the CPU to memory or I/O devices. This is because the CPU is the component responsible for specifying the address to either read from or write to.

2. Width (Bus Width)

- The width of the address bus, measured in bits, determines how many memory locations the system can access. For example:
 - > A 32-bit address bus can access 2^{32} memory locations (approximately 4 GB of memory).
 - > A 64-bit address bus can access 2^{64} memory locations (allowing access to much larger amounts of memory).

3. Role in the System

- When the CPU needs to read or write data,

it first sends the address of the data location via the address bus to memory or an I/O device. Once the address is specified, the data can be transferred using the data bus.

* Control Bus

→ The control Bus is a part of a computer's system bus that carries control signals from the CPU to other components of the computer, such as memory and (I/O) input/output device. It is responsible for co-ordinating and managing the actions of the system, ensuring that all components work together efficiently by sending control signals that tell the components ~~what~~ to do.

• Key Features of the Control Bus:

① Direction

- The control bus can be unidirectional or bi-directional, depending on the specific control signals. Some signals are sent from the CPU to other components, while others can come back from components to the CPU (e.g., interrupt signals).

② Control signals

- The control bus carries various types of

control signals that manage the operations of the computer. Some of the key control signals include:

- > Read / write signals
- > ~~the~~ clock signals
- > Interrupt Request (IRQ)
- > Bus Request / Grant signals
- > Memory or I/O select.

Assignment - 3

Difference between SRAM and DRAM

SRAM	DRAM
① It is known as static Random Access Memory.	It is known as Dynamic Random Access Memory.
② Data is stored using flip-flops made of about six transistors for each bit.	Data is stored using one transistor and one capacitor for each bit.
③ It does not require refreshing as long as continuous power is supplied.	It requires periodic refreshing because the charge in capacitor leaks away.
④ It is very fast in operation and has a low access time.	It is slower in speed compared to SRAM and has higher access time.

- | | |
|---|--|
| ⑤ It is more costly because of its complex circuit design. | It is cheaper because of its simpler circuit design. |
| ⑥ SRAM are commonly used in cache memory and CPU registers. | DRAM are commonly used as main memory (RAM) in computers. |
| ⑦ It has lower density, so it stores less data in the same physical area. | It has higher density, so it stores more data in the same physical area. |

Difference between Mainframe and Super Computer.

Mainframe Computer	Super Computer
① Designed to handle thousands of users and process huge amounts of data simultaneously.	Designed to perform extremely fast calculations and solve complex scientific problems.
② Mainly used for business applications like banking, insurance, and transaction processing.	Mainly used for research applications like weather forecasting, space research, nuclear simulation, and AI.

③ Focuses on handling large data input / output operations.	Focuses on high-speed mathematical calculations.
④ Slower in processing speed compared to supercomputers.	World's fastest computers with speeds measured in petaflops or exaflops.
⑤ Very large in size but smaller compared to super computers.	Extremely large in size, often covering entire rooms.
⑥ Supports thousands of users working at the same time	Usually dedicated to a single task or limited group of researchers.
⑦ eg: IBM Zseries, Hitachi Z800	eg: Summit, Fugaku Frontier.

Explain about Magnetic drum in brief.

→ A magnetic drum is an early form of computer memory that was widely used in the 1950s and 1960s. It is a metal cylinder coated with a magnetic material, and data is stored on its surface in the form of magnetic patterns. The drum ~~was~~ rotates at high

speed, and fixed read/write heads placed along its length are used to access the stored data.

Magnetic drums were mainly used as the main memory in first generation computers and later as secondary storage. Although they are now obsolete they were important in the ~~high~~ history of computer memory technology.

Write 10 differences between SSD and HDD

SSD	HDD
1) It uses flash memory to store data.	It uses spinning magnetic disks to store data.
2) SSD has much faster data read and write speeds.	HDD has slower data read and write speed.
3) SSD has ^{no} mechanical moving parts.	HDD has moving parts like spinning disks and a read/write head.
4) SSD is more resistant to shock and physical damage.	HDD is more likely to be damaged by drops or shocks.
5) Consumes It consumes less electrical power.	It consumes more electrical power.
6) It produces less heat during operation.	It produces more heat during operation.
7) SSD is more expensive for the same amount of storage.	HDD is cheaper for the same amount of storage.

8) SSD is usually smaller in size and lighter in weight.	HDD is larger in size and heavier in weight.
9) SSD has limited write cycles but long life in normal uses.	HDD suffers from wear and tear of mechanical parts.
10) SSD works silently.	HDD produces noise from spinning disks.

Explain BCR, OCR, OMR, MICR in Brief.

1. Barcode Reader (BCR)

↳ A Barcode Reader (BCR) is an input device that reads information from barcodes printed on products, books, or inventory items. It converts the scanned data into digital form and sends it to the computer, making tasks like billing, stock management, and product identification fast and accurate.

2. Optical Character Recognition (OCR)

↳ OCR is a device or software that reads printed or typed text from documents and converts it into digital form. It is widely used for scanning books, forms, and papers to make the text editable and searchable on a computer.

3. Optical Mark Recognition (OMR)

→ OMR detects marks made on paper, such as tick marks or filled bubbles. It is commonly used in multiple-choice exams, surveys, and polls to collect data quickly and accurately without manual entry.

4. Magnetic Ink character Recognition (MICR)

→ MICR reads characters printed with special magnetic ink, usually found on bank cheques. This input device allows fast and secure processing of financial transactions, reducing errors and improving efficiency in banks.

Explain printer with its types and 2-2 advantages & disadvantages of each.

→ An output device that produces a hard copy of digital documents, images, or graphics on paper is called Printer. It converts soft copy (from computer) into physical form. Printers can be of various types depending on technology and speed. There are mainly two types of printer.

> Impact printer

- Impact printers work by striking a print head against an ink ribbon, which then tr-

transfers the ink onto the paper.

Types of Impact printer

① Dot matrix Printer

- Dot matrix printers use a print head that moves back and forth striking an ink ribbon against the paper to form dots that together create characters.
- Advantage
 - Cost effective
 - Durable and reliable
- Disadvantage
 - Low print quality
 - Noisy operation

② Daisy wheel Printer

- Daisy wheel printers use a wheel with characters on the tips of its spokes. The wheel spins to the required character and strikes it against the ink ribbon to print on paper.
- Advantage
 - High-quality text printing
 - Faster than dot matrix printers for text.
- Disadvantage
 - Noisy
 - Cannot print graphics (only text)

③ Line Printer

- A type of impact printer that print an entire line of text at once, rather than character by character is known as line printer.

- Advantage
 - High durability and reliability
 - Very fast for printing large volume of text.
- Disadvantage
 - Very noisy
 - Bulky and expensive

> Non Impact Printers

> Non Impact Printers print without striking paper; use lasers, inkjets, or thermal method.

Types of Non Impact Printers

① Ink-jet Printers

- Ink-jet Printers are non-impact printers that work by spraying tiny droplets of liquid ink directly onto paper through microscopic nozzles.

- Advantage
 - Very high resolution (up to 1200-4800 dpi or more)
 - High print quality with sharp text and detailed graphics.

• Disadvantage

- High running cost due to expensive ink cartridges.
- Slower printing speed compared to Laser printers.

② Laser printer

- A non-impact printer that uses a laser beam and electro-photographic technology to produce high-quality text and graphics is known as laser printer.

• Advantage

- High printing speed (20-60 pages per minute)
- Economical for bulk / high-volume printing (low cost per page)

• Disadvantage

- High initial purchase cost
- Needs warm-up time before printing

③ Thermal printing

- A type of non-impact printer that produces images on paper by selectively heating coated thermal paper or by melting wax/ink from a ribbon onto the paper.

• Advantage

- High speed (5-12 inch per second)
- Cost effective (low cost)

• Disadvantage

- Monochrome
- Not for Graphics.

Write down 8 differences between CRT and LCD monitor.

CRT Monitor	LCD Monitor
① It works by firing electron beams onto a phosphor-coated screen inside a large vacuum tube.	Uses liquid crystals between two polarized glass panels with a backlight to produce images.
② It requires a lot of desk space and is difficult to move because it is very bulky, large & heavy.	It takes little desk space, easy to carry because it is slim, flat & light weight.
③ Consumes more electricity since it needs high voltage to accelerate electrons for image formation.	Consumes less power because it only needs a backlight for illumination.
④ Image quality is lower and often suffers from flickering, which may cause eye strain during long use.	Provides stable, flicker-free, sharp, and clear images, comfortable for extended viewing.
⑤ CRT Monitor's screen can be curved or	LCD Monitor to can have perfectly

flat, but often shows geometric distortions, especially at the edges.	flat screen with no distortion, giving accurate geometry and a modern look.
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⑥ It can display multiple resolutions but sharpness reduces at higher settings; text may look blurry.	It can be Optimized for a 'native resolution' offering crystal-clear images & supporting HD graphics.
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⑦ It Generates more heat and emit radiation due to electron beam activity; less environmentally friendly.	It It generates very little heat and emits negligible radiation, making it safer and eco-friendly.
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⑧ It has shorter lifespan because phosphor coating burns out over time.	It has longer lifespan, more reliable, durable and consistent in performance for many years.
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Explain PS/2 ports in brief.

→ A PS/2 port is a round 6-pin connector that was introduced by IBM in 1987 with its personal system /2 computers. It was mainly used to connect a keyboard and mouse to a desktop computer. The ports are color coded for easy identification, with purple for keyboard and green for the mouse. PS/2 uses a serial communication protocol, which is slower compared to modern USB connections, but it provided stable and reliable input for many years.